Computer Communication Laboratory Seminar

Satisfaction-Aware Resource Allocation for Integrated OIRS-aided HAP and sub-THz Access Networks

Student: DANG Dinh Khanh – m5282002

Aizuwakamatsu, June 17, 2026

Outline of Presentation

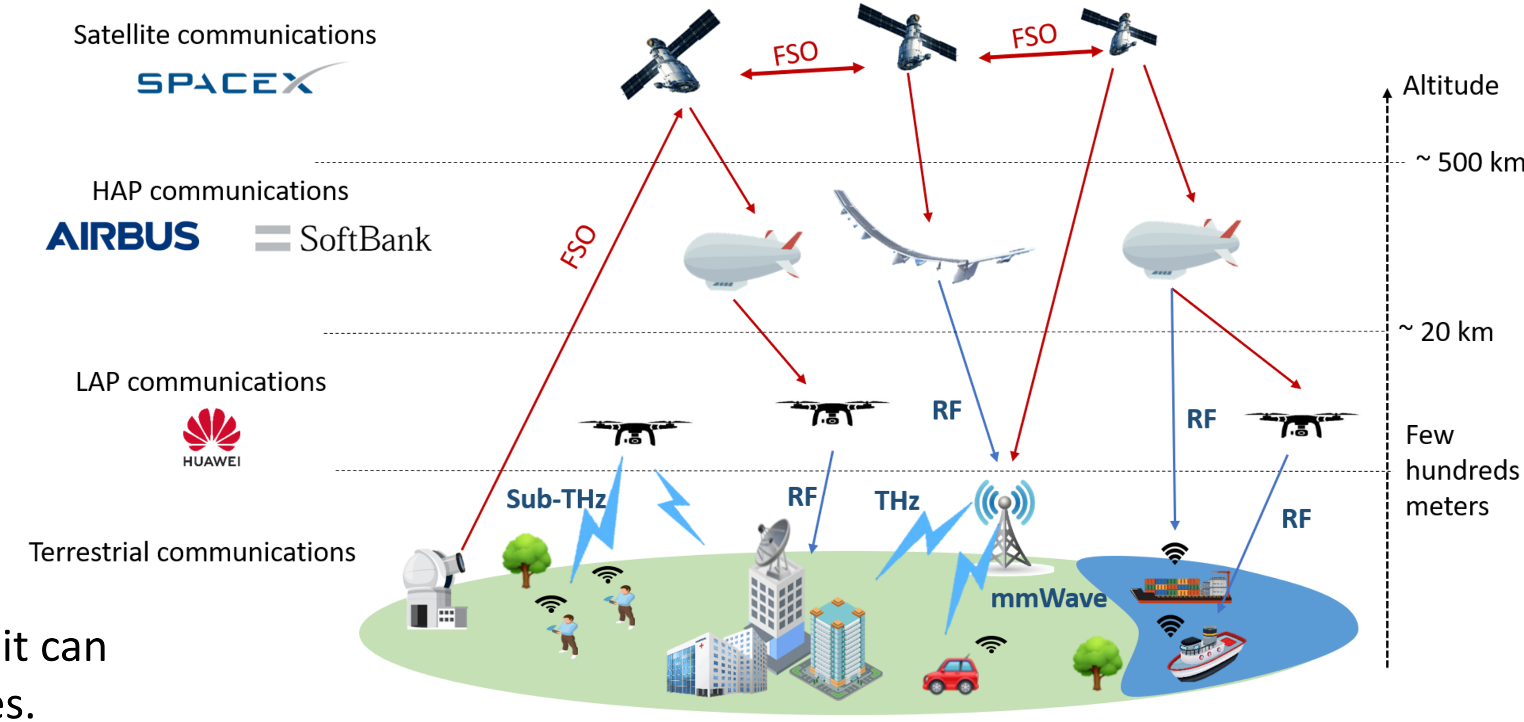
1. Research Background: 6G Vision for Network Architectures and Communication Technologies

- Future 6G networks are expected to support **ubiquitous connectivity** and **high-data services**, even in remote, rural, and hard-to-reach areas.

- Non-terrestrial networks (NTNs) *using satellites, HAPs and UAVs* can *extend service coverage beyond terrestrial infrastructure,* especially useful for connecting rural and remote areas.

- FSO backhaul (GS-> NTNs-> UAV):
  - Use near-infrared light to transmit data
  - Extremely high data rate (Gbps to Tbps).

- Access (UAV/BS -> end users):
  - Incorporate RF/mmWave/THz/sub-THz
  - Sub-THz access is recently attracted because it can provide large bandwidth for high-rate services.



➡ **The integration of FSO backhaul-based NTNs and sub-THz access can provide large coverage and high-data-rate service.**

1. Research Background: The Optical IRS-assisted FSO-based NTNs

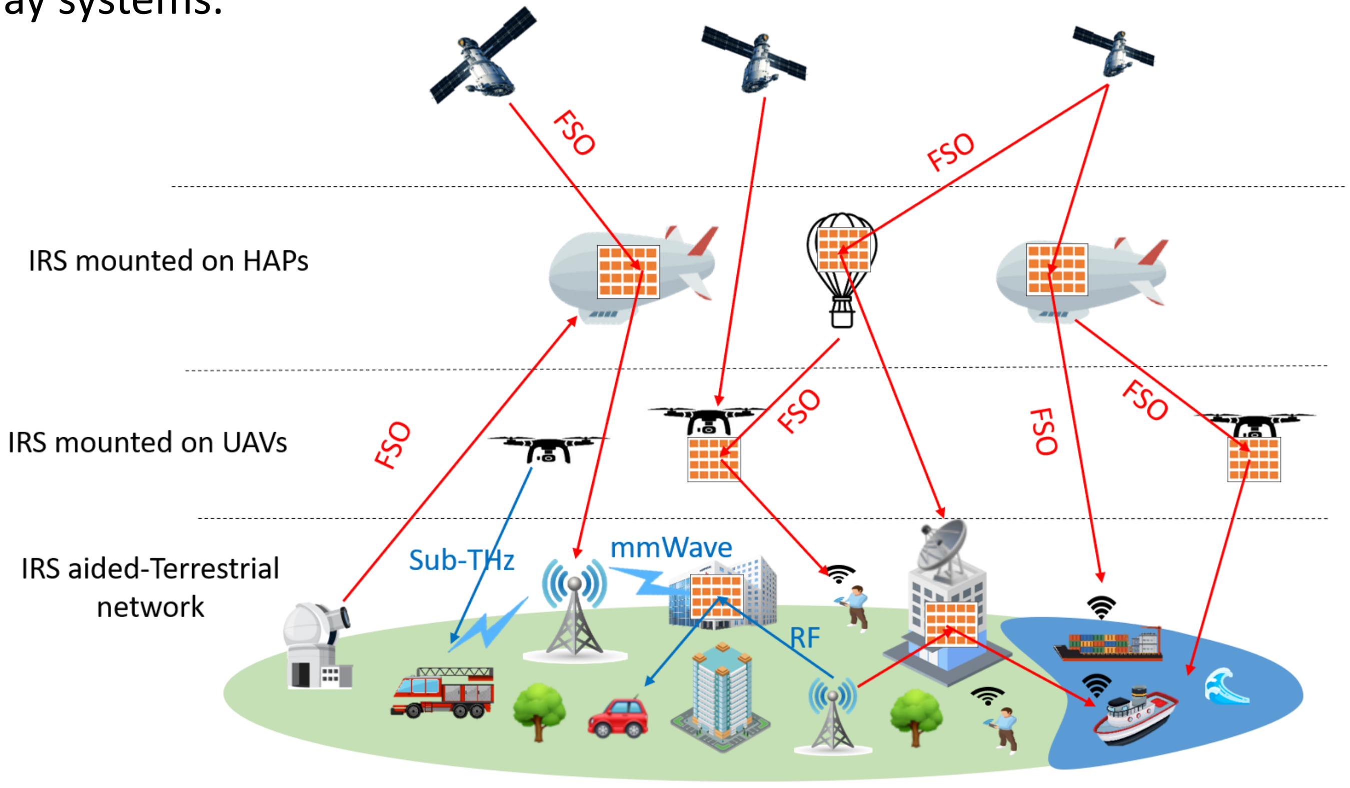
- FSO backhaul is attractive for high-capacity aerial networks, but relay-based FSO systems have strict *limits on cost size, weight, and power consumption (C-SWaP) and hardware complexity.*
- To relax this limitation, many studies have developed ***optical intelligent reflecting surfaces (OIRS)*** as a potential technology to *serve as a lightweight alternative* to conventional relay systems.

Optical Intelligent Reflecting Surface (OIRS)

- A passive surface with mirrors or meta-atoms ***reflects FSO beam*** in a ***controlled way.***
- Possible to simultaneously support multiple users

Advantage compared to relay systems

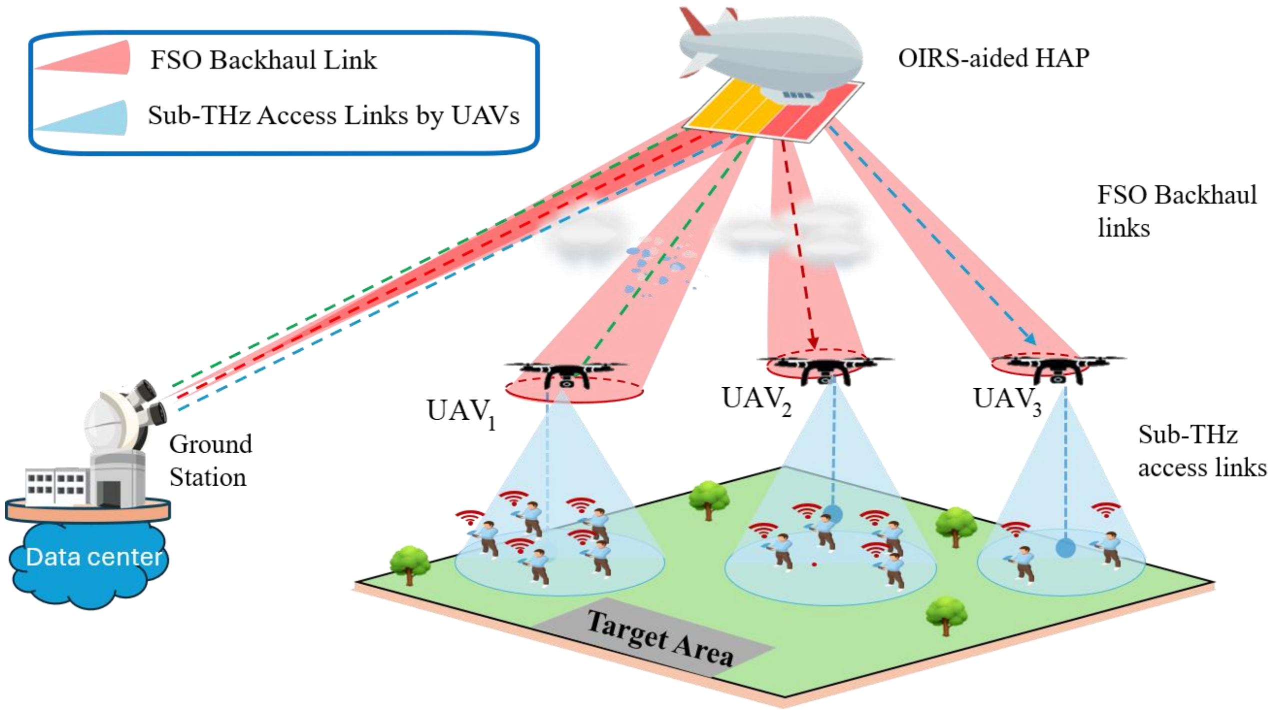
- Lightweight, low cost
- Low power consumption
- Extend coverage, avoid blockage



The operation concept of optical IRS

The HAP-mounted OIRS can assist multiple UAVs backhaul links by allocating OIRS blocks, enabling flexible high-capacity FSO backhaul.

2. System Model: Integrated OIRS-aided HAP and sub-THz Access network



Integrated OIRS-aided HAP and Sub-THz Access Networks

FSO Backhaul side

Ground station –> HAP-mounted OIRS –> *N fixed UAVs*

The OIRS comprises N_B blocks for allocation

Sub-THz Access side

Each UAV acts as an aerial hotspot base station serving ground users.

UAV hotspots are assumed to be spatially separated

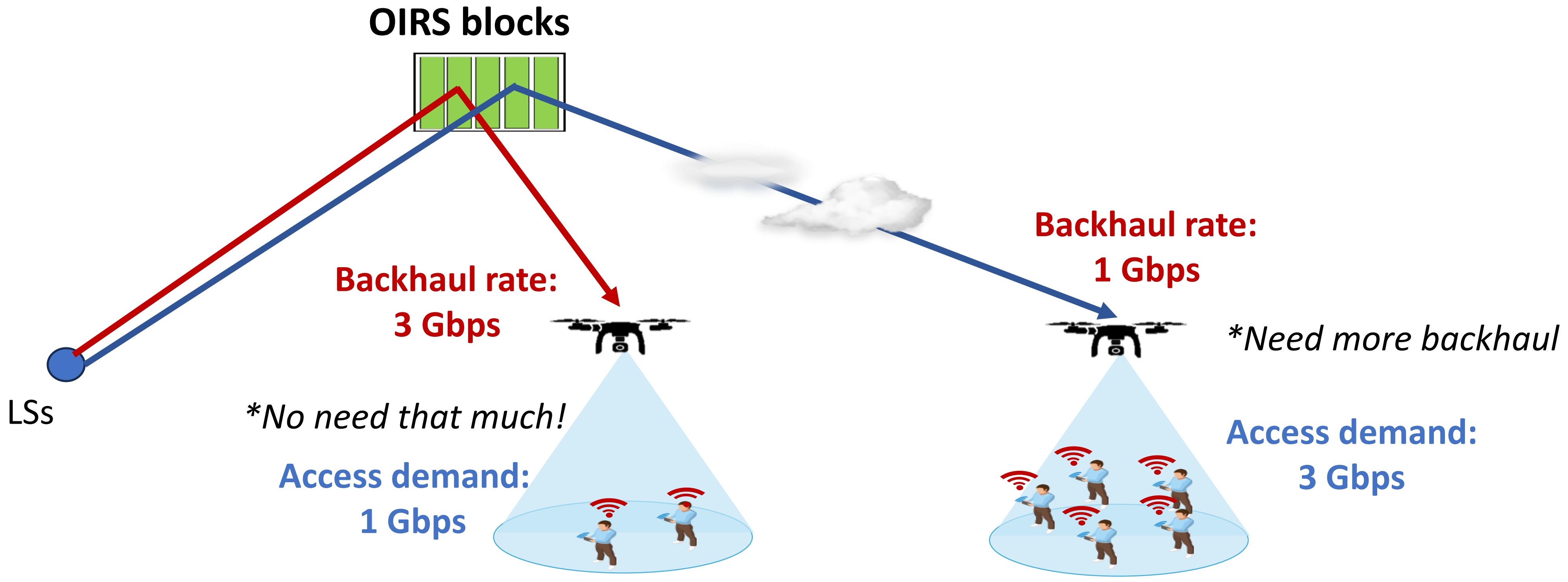
End-to-end service

Ground users request different services (VR/web browser/download app…)

The HAP/OIRS allocates UAV-level backhaul capacity, while each UAV performs user-level sub-THz scheduling.

❑ 3.1 Access-Backhaul Traffic Mismatch

- Due to C-SWaP, **OIRS blocks** mounted on HAP are **limited and shared among *N* UAVs**

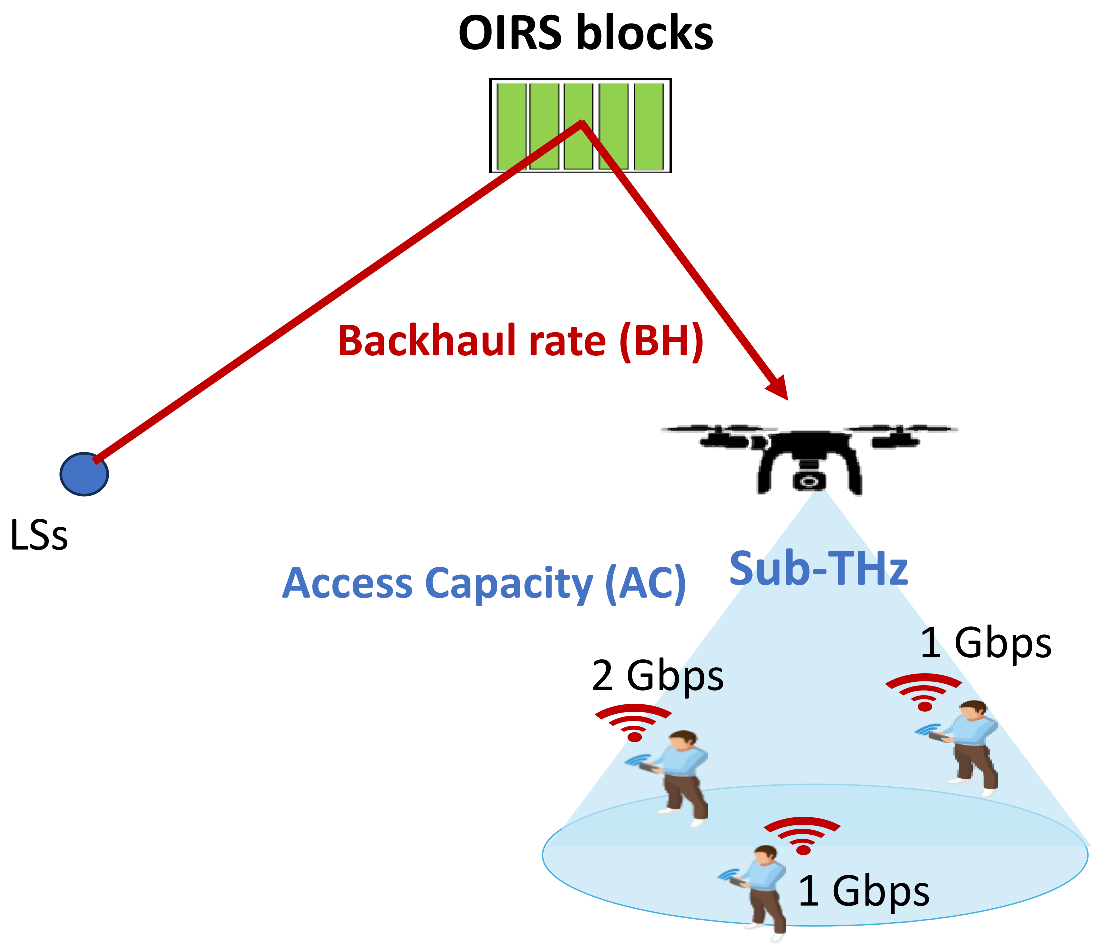


Low-demand case: additional OIRS blocks may be wasted

Backhaul bottleneck (insufficient rate): more OIRS blocks are needed

➡ **OIRS allocation should consider both *backhaul channel condition* and *UAV-side access demand/capability*.**

❑ 3.2 Limited Power and Bandwidth in sub-THz Access



❖ Each UAV has limited power and available bandwidth

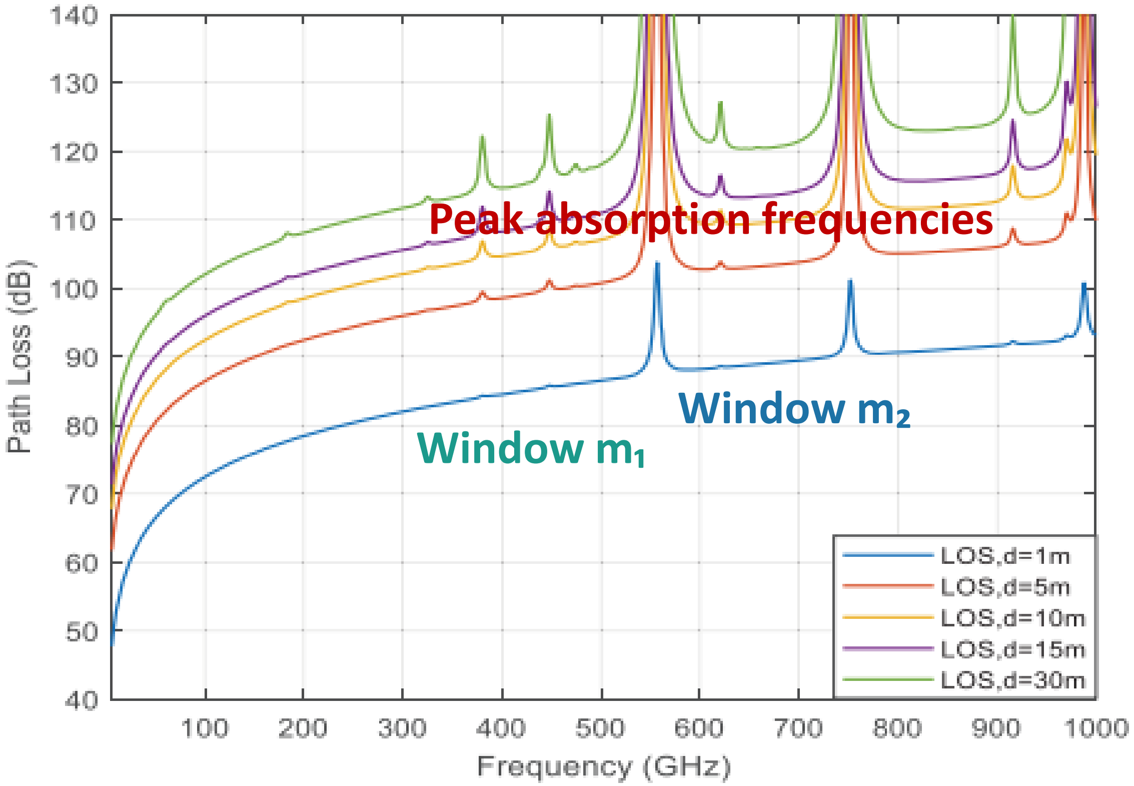
- Power and bandwidth should be allocated to users according to demand, channel condition, and available backhaul.
- If backhaul-limited case: access capacity > backhaul capacity,
  - Example: BH 2Gbps, AC 4Gbps, Demand 4Gbps
    -> Using full bandwidth and power may be wasted
- If access-limited case: access capacity is not sufficient
  - Example: BH 4Gbps, AC 2Gbps, Demand 4Gbps
    -> more backhaul cannot increase served traffic
    -> bandwidth and power need to be allocated to users

Power, bandwidth allocation and OIRS allocation should be jointly designed with user demands

❑ 3.3 Sub-THz Impairments

- In sub-THz access links, the achievable rate strongly depends on user distance, channel condition, and selected frequency window.

  - Beam spreading loss: weaker signal by distance
  - Small-scale fading: instantaneous channel gains
  - Molecular absorption: air molecules absorb signal energy



Window m_1
Lower absorption
Suitable for long-distance users
Possibly limited usable bandwidth

→

Window m_2
Higher frequency
Potentially larger bandwidth
More sensitive to distance

→

Decision
Allocate frequency windows based on user distance, channel, and demand

❑ 3.4 Dynamic Users and Historical Satisfaction State

- Due to user mobility, time-varying channel, and limited resources, some users **may be unserved (low satisfaction)** repeatedly.

  - **End-to-end demand satisfaction of user *k*** associated with UAVi:

$$\rho_{i,k}(t) = \frac{\text{Served rate}}{\text{Demand}} = \frac{s_{i,k}(t)}{D_{i,k}^{\text{req}}(t) + \delta}, \qquad 0 \leq \rho_{i,k}(t) \leq 1.$$

- $\rho_{i,k}(t) = 1$, the demand of user k is fully served.
- $\rho_{i,k}(t)$ is close to 0, only a small portion of the requested traffic is served.

  - Historical satisfaction is used to record past service quality.
  - Users with low historical satisfaction receive larger priority weights in later slots.

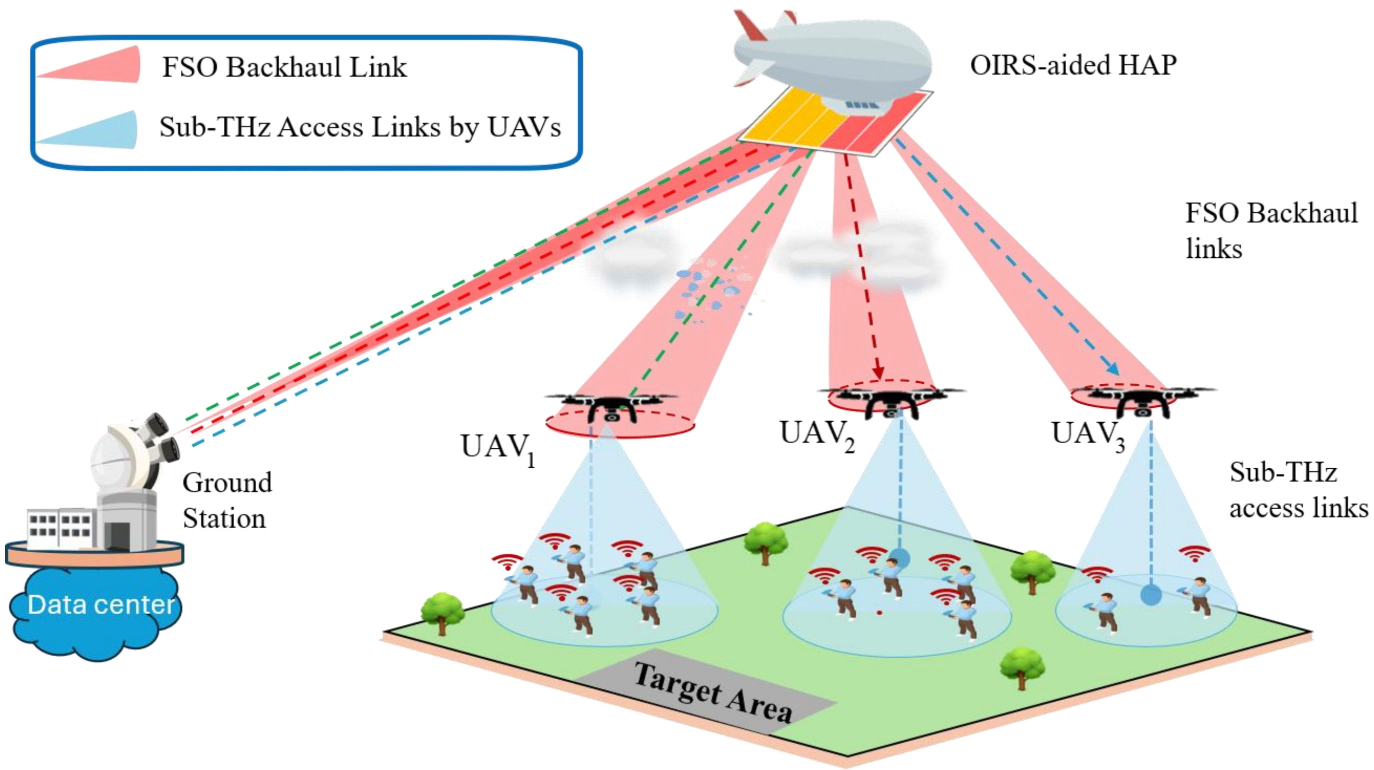
-> Historical satisfaction state and priority weight increase the chance that repeatedly underserved users receive resources in later slots

3. Research Gaps and Goals

Refs	System Model	Main Objective	OIRS-FSO Backhaul	Integrated Access/Backhaul	sub-THz Access	Traffic-Aware Demand Satisfaction	Moving Cloud/ Multi-HAP
UAV RF access [1]-[3]	UAV-assisted RF access	UAV placement / trajectory to maximize covered users, throughput, or user satisfaction	✗	△	✗	△	✗
OIRS-assisted FSO backhaul [4]	OIRS-assisted FSO backhaul	Improve FSO capacity, outage, or link reliability	✓	✗	✗	✗	△
OIRS allocation Backhaul [5]	HAP-OIRS-UAV FSO backhaul	Maximize number of supported UAVs under fixed target backhaul rate	✓	✗	✗	✗	✓
Integrated FSO/RF access [6]-[10]	UAV IAB with RF/mmWave access	Joint UAV placement / trajectory / bandwidth allocation	△	✓	✗	△	△
OIRS-HAP + UAV RF access under clouds [11]-[12]	HAP-OIRS FSO backhaul + UAV RF access	Maximize number of supported users under dynamic users/clouds	✓	✓	✗	△	✓
THz/sub-THz access networks [13]-[14]	THz/sub-THz Base Station/UAV	Transmission-window selection, sub-band assignment, power allocation, user association	✗	△	✓	△	✗
This work	**HAP-OIRS FSO backhaul + UAV sub-THz access**	**Maximize weighted end-to-end user satisfaction**	✓	✓	✓	✓	**Future extension: ✓**

✓: explicitly considered　　△: partially considered /simplified　　✗: not considered

This work designs *a satisfaction-aware resource allocation scheme* that jointly considers *OIRS block allocation, sub-THz power/bandwidth/frequency-window allocation, and historical user satisfaction* to *improve end-to-end users' satisfaction.*



- Number of users associated with UAVi

$$K_i(t) \sim \text{Poisson}(\lambda_i(t))$$

- Per-user requested traffic demand

$$d_{i,k}(t) \in \{d_{\text{IoT}}, d_{\text{video}}, d_{\text{eMBB}}, d_{\text{emergency}}\}$$

- Backhaul rate with *n* allocated blocks

$$R_i^{\text{BH}}(n_i;t) = B_{\text{FSO}} \log_2 \left(1 + \gamma_i^{\text{FSO}}(n_i;t)\right)$$

- The achievable sub-THz access rate of user k associated with UAVi, given allocated power p_i and bandwidth b_i

$$R_{i,k}^{\text{AC}}(\mathbf{p}_i, \mathbf{b}_i; t) = \sum_{m \in \mathcal{M}} b_{i,k,m}(t) \times \log_2 \left(1 + \frac{p_{i,k,m}(t) g_{i,k,m}(t)}{b_{i,k,m}(t)\left[N_0 + N_{\text{mol}}(f_m, d_{i,k}(t))\right]}\right),$$

- f_m is the center frequency of window m,
- N_0 is the thermal noise power spectral density, $N_{\text{mol}}(f_m, d_{i,k}(t))$ denotes the molecular absorption noise over the sub-THz link,
- $g_{i,k,m}(t)$ is the channel gain.

4. Proposed Resource Allocation: Served Traffic and Utility

❑ End-to-end served traffic and satisfaction ratio:

- We define the actual end-to-end served rate of user k associated with UAVi

$$s_{i,k}(t) = \min(R^{\mathrm{AC}}_{i,k}(\mathbf{p}_i, \mathbf{b}_i; t), D^{\mathrm{req}}_{i,k}(t))$$

- Total end-to-end served rate of all users associated with UAVi

$$\sum_{k \in \mathcal{K}_i(t)} s_{i,k}(t) \le R^{\mathrm{BH}}_i(n_i; t),$$

- ***End-to-end demand satisfaction of user k*** associated with UAVi:

$$\rho_{i,k}(t) = \frac{\text{Served rate}}{\text{Demand}} = \frac{s_{i,k}(t)}{D^{\mathrm{req}}_{i,k}(t) + \delta},$$

- ***Per-user satisfaction-aware utility:***

$$u_{i,k}(t) = \ln\left(1 + \rho_{i,k}(t)\right).$$

4. Proposed Resource Allocation: Historical Satisfaction and Priority Weight

- ***For each user k associated with UAVi, we consider not only current satisfaction but also past service quality***
  - The *target satisfaction is* ρ^{tar}
  - The historical satisfaction state is updated by: $\bar{\rho}_{i,k}(t+1) = (1-\alpha)\bar{\rho}_{i,k}(t) + \alpha\rho_{i,k}(t),$
  - The satisfaction deficit is $d_{i,k}^{\text{his}}(t) = \left[\rho^{\text{tar}} - \bar{\rho}_{i,k}(t)\right]^{+},$
  - ***The normalized historical priority weight of each user k associated with UAVi:***

$$w_{i,k}(t) = K_{\text{tot}}(t)\frac{1+\eta d_{i,k}^{\text{his}}(t)}{\sum\limits_{j\in\mathcal{I}}\sum\limits_{\ell\in\mathcal{K}_j(t)}\left[1+\eta d_{j,\ell}^{\text{his}}(t)\right]},$$

- α: controls how much the current satisfaction affects the historical state.
- η: controls how strongly historical deficit increases priority.
- Newly active users are initialized with $\bar{\rho}_{i,k}(t) = \rho^{tar}$

-> By incorporating per-user utility function and priority weights, we can formulate objective problem

Optimization Problem:

$$\mathbf{P1}: \max_{n,p,b,\rho} \sum_{i\in\mathcal{I}} \sum_{k\in\mathcal{K}_i(t)} w_{i,k}(t) \ln\left(1+\rho_{i,k}(t)\right)$$

Maximize weighted end-to-end satisfaction

$$\text{s.t.} \quad \rho_{i,k}(t) = \frac{s_{i,k}(t)}{D^{\text{req}}_{i,k}(t)+\delta}, \quad \forall i \in \mathcal{I},\ k \in \mathcal{K}_i(t),$$

$$0 \le s_{i,k}(t) \le D^{\text{req}}_{i,k}(t), \quad \forall i \in \mathcal{I},\ k \in \mathcal{K}_i(t),$$

$$s_{i,k}(t) \le R^{\text{AC}}_{i,k}(\mathbf{p}_i, \mathbf{b}_i; t), \quad \forall i \in \mathcal{I},\ k \in \mathcal{K}_i(t),$$

Served traffic constraints

$$\sum_{k\in\mathcal{K}_i(t)} s_{i,k}(t) \le R^{\text{BH}}_i(n_i; t), \quad \forall i \in \mathcal{I},$$

Backhaul constraint

$$\sum_{k\in\mathcal{K}_i(t)} \sum_{m\in\mathcal{M}} p_{i,k,m}(t) \le P^{\text{U}}_i, \quad \forall i \in \mathcal{I},$$

UAV power constraints

$$\sum_{k\in\mathcal{K}_i(t)} b_{i,k,m}(t) \le B_m, \quad \forall i \in \mathcal{I},\ m \in \mathcal{M},$$

UAV bandwidth constraints

$$\sum_{i\in\mathcal{I}} n_i(t) \le N_B,$$

OIRS block budget

$$p_{i,k,m}(t) \ge 0, \quad b_{i,k,m}(t) \ge 0, \quad s_{i,k}(t) \ge 0,$$

$$n_i(t) \in \mathbb{Z}_+, \quad \forall i \in \mathcal{I}.$$

Algorithm: Hierarchical Value-Function-Based Allocation

- Key idea: allocate each OIRS block to UAV where additional backhaul can ***most effectively improve weighted user satisfaction***.
- 2 levels optimization: HAP-level (dynamic programming) + local UAV scheduling (convex optimization)

How the system behaves:

- **Behavior 1:** backhaul-limited UAV needs more backhaul to serve access demand
  -> allocate more blocks
- **Behavior 2:** access-limited UAV, giving more backhaul is not very useful
  -> allocate less blocks
- **Behavior 3:** low-satisfaction user are favored
  -> small resource can gain high satisfaction
- **Behavior 4:** historically underserved user receive larger priority weights
  -> Historical weights increase the priority of users served poorly in previous slots.

Algorithm 1 Hierarchical Value-Function-Based OIRS Allocation

```
Input: Active user sets {K_i(t)}, user demands {D^req_{i,k}(t)}, sub-THz channel states {g_{i,k,m}(t)}, backhaul rate functions {R_i^BH(q; t)}, historical states {ρ̄_{i,k}(t)}, and total OIRS blocks N_B
Compute historical weights {w_{i,k}(t)} according to (56)–(57)
for each UAV i ∈ I do
    for q = 0 to N_B do
        Solve the local UAV problem (62) with q OIRS blocks
        Store the obtained value F_i(q; t)
    end for
end for
Initialize V(0, ℓ) = 0 for all ℓ = 0, ..., N_B
for i = 1 to |I| do
    for ℓ = 0 to N_B do
        V(i, ℓ) = max_{0≤q≤ℓ} {V(i − 1, ℓ − q) + F_i(q; t)}
        Store the selected q for backtracking
    end for
end for
Obtain the optimal OIRS block allocation {n_i^⋆(t)} by backtracking
Apply the corresponding local access allocation {p_i^⋆, b_i^⋆, s_i^⋆}
for each UAV i ∈ I do
    for each active user k ∈ K_i(t) do
        Compute ρ_{i,k}(t) using (52)
        Update ρ̄_{i,k}(t + 1) using (54)
    end for
end for
Output: OIRS block allocation {n_i^⋆(t)} and UAV-side access scheduling variables {p_i^⋆, b_i^⋆, s_i^⋆}
```

4. Proposed Resource Allocation: Protocol Flow

1. Dynamic Users and Hotspot State
Start a timeslot
Observe active users, locations and demands

2. Access and Backhaul Channel Observation
Estimate sub-THz access channels and FSO backhaul gain

3. Historical priority computation
Update historical satisfaction state
Calculate deficit and priority weight of each user

4. Local UAV computation
For each possible OIRS blocks, solve the local user scheduling and obtain a utility value
-> A set of utilities correspond to allocated OIRS blocks

5. HAP/OIRS allocation
Based on the set of utility in step 4, HAP allocates OIRS blocks to UAVs to maximize total satisfaction

6. UAV-side sub-THz scheduling
After OIRS allocation, allocate power/bandwidth and frequency-window

5. Performance Metrics and Benchmarks

Performance metrics:

- Total end-user satisfaction utility: $$U_{\text{total}}(t) = \sum_{i \in \mathcal{I}} \sum_{k \in \mathcal{K}_i(t)} w_{i,k}(t) \ln\left(1 + \rho_{i,k}(t)\right).$$
- Average end-user satisfaction: $$\rho_{\text{avg}}(t) = \frac{1}{K_{\text{tot}}(t)} \sum_{i \in \mathcal{I}} \sum_{k \in \mathcal{K}_i(t)} \rho_{i,k}(t).$$
- Total served rate: $$S_{\text{total}}(t) = \sum_{i \in \mathcal{I}} \sum_{k \in \mathcal{K}_i(t)} s_{i,k}(t).$$
- Satisfied-user ratio: $$\Gamma(t) = \frac{K_{\text{sat}}(t)}{K_{\text{tot}}(t)}.$$

To evaluate our proposed scheme, we compare it with ***two benchmarks***:

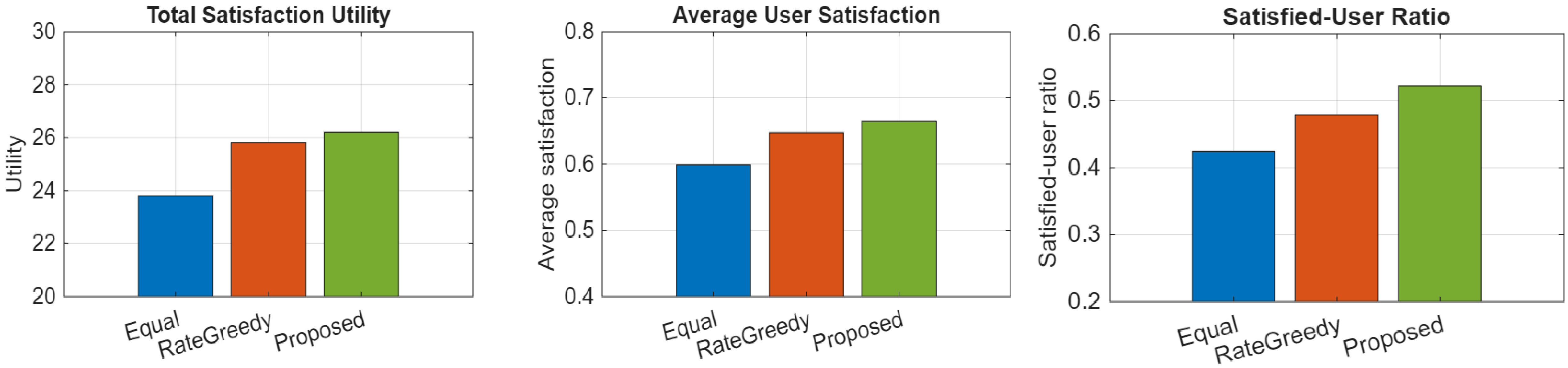
(1) Rate-based Greedy Allocation (RGA): allocate OIRS according to the highest marginal served-rate gain.

(2) Uniform Element Allocation (UEA): the OIRS is shared equally among all UAVs.

Parameters: 5 UAVs with [15, 5, 12, 8, 10] users, respectively.

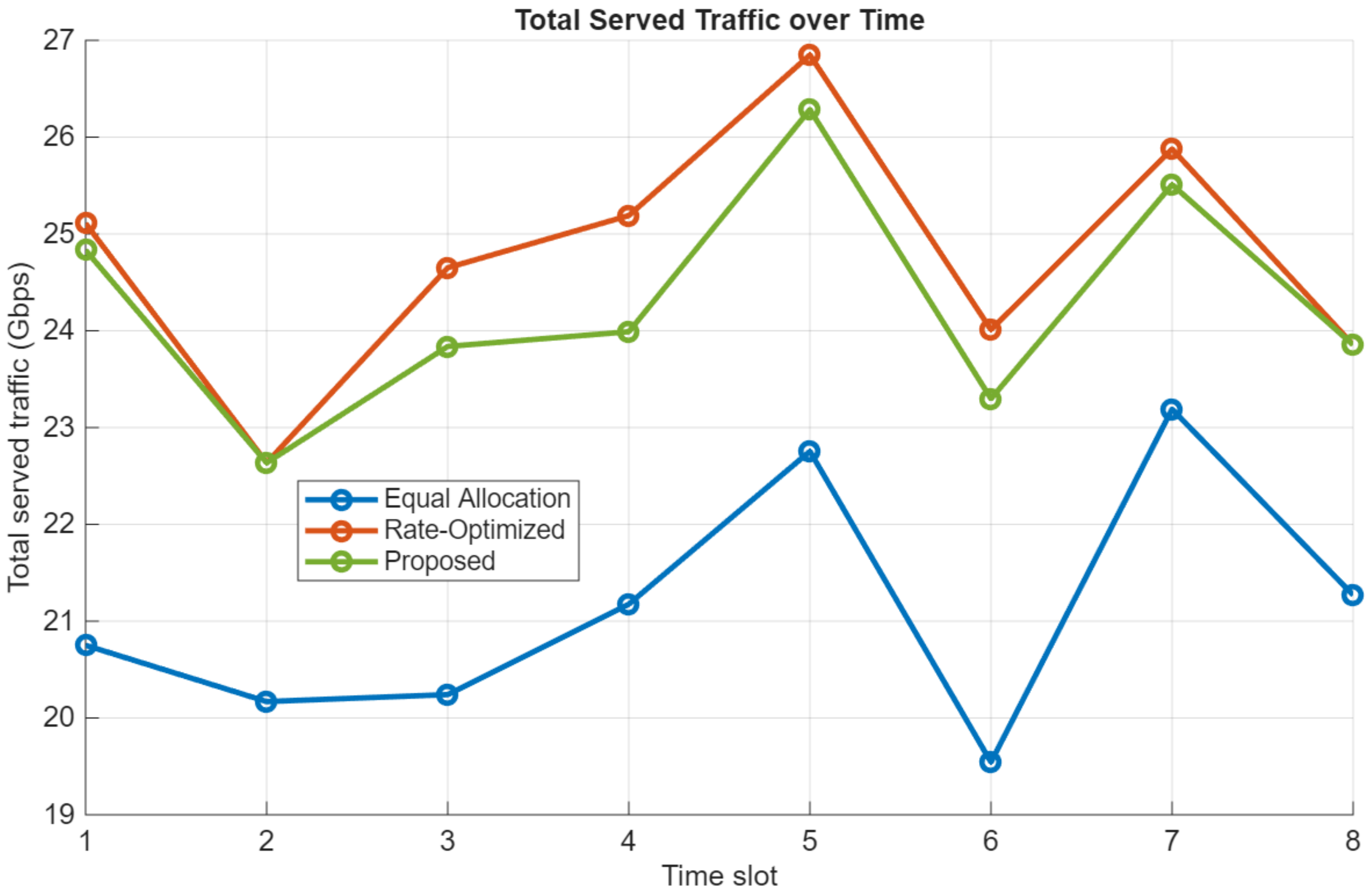
- Service rate classes of users = [100 Mbps, 600Mbps, 1Gbps, 2Gbps];
- Cloud liquid water content = [0.5, 2, 1.5, 1.5, 0.5];

5. The Effectiveness of the Proposed Satisfaction-Aware Resource Allocation



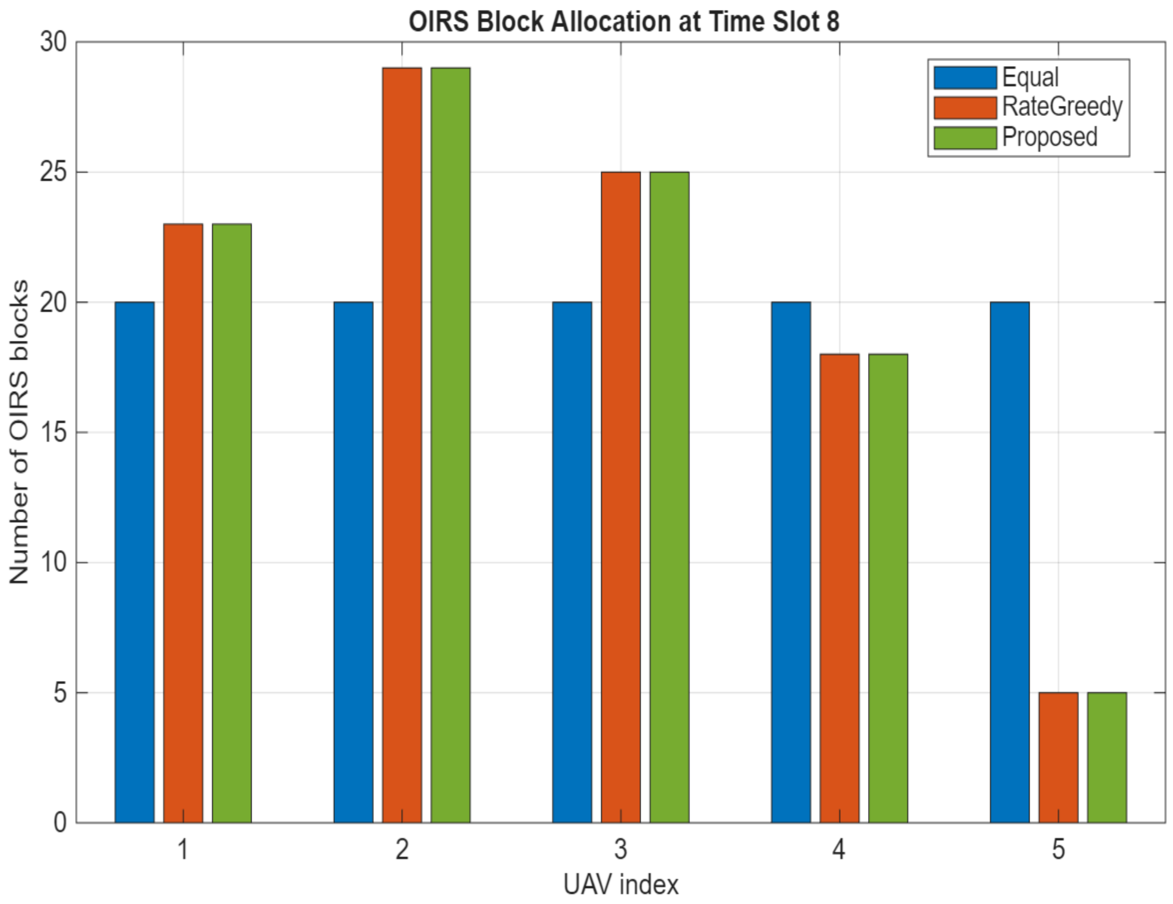
- The proposed allocation ***outperforms other schemes by:***
    - ***Highest total satisfaction***
    - ***Average user satisfaction is improved***
    - ***Number of satisfied users (user achieves target satisfaction) is increased***

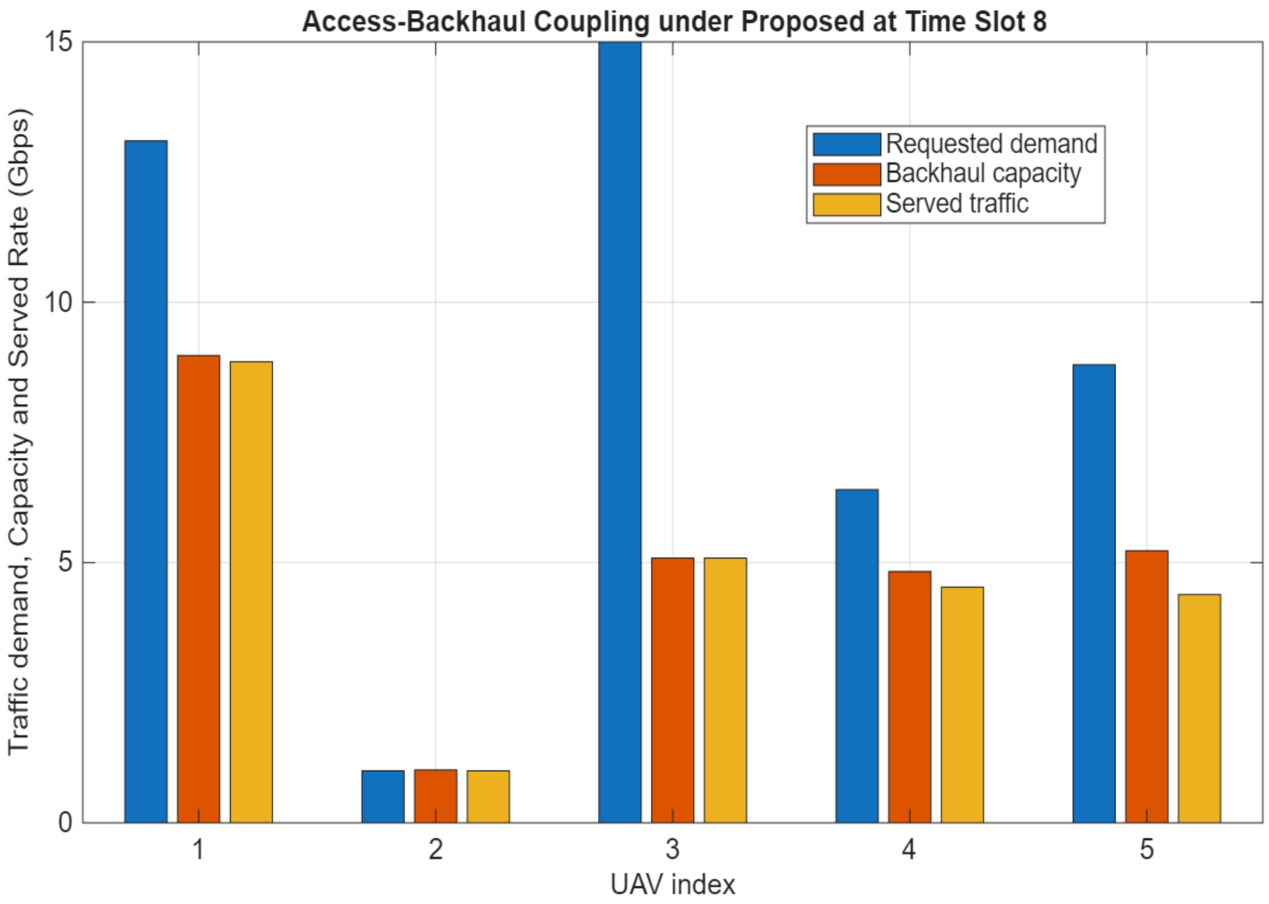
5. The Rate-Satisfaction Tradeoff



- Although the proposed allocation ***can provide higher total satisfaction, the total served rate is not optimized.***

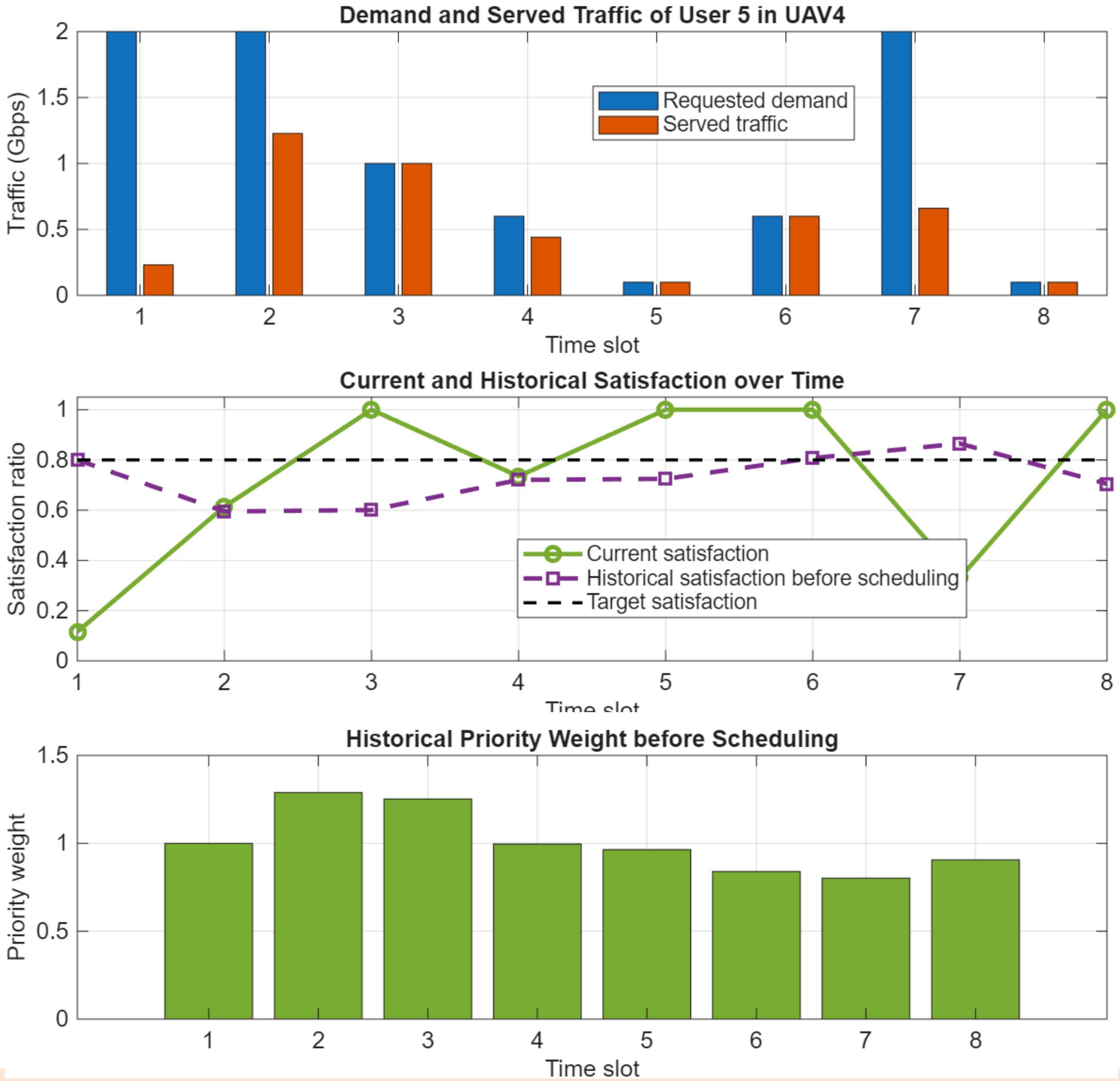
5. OIRS Allocation Behavior in the Proposed Satisfaction-Aware





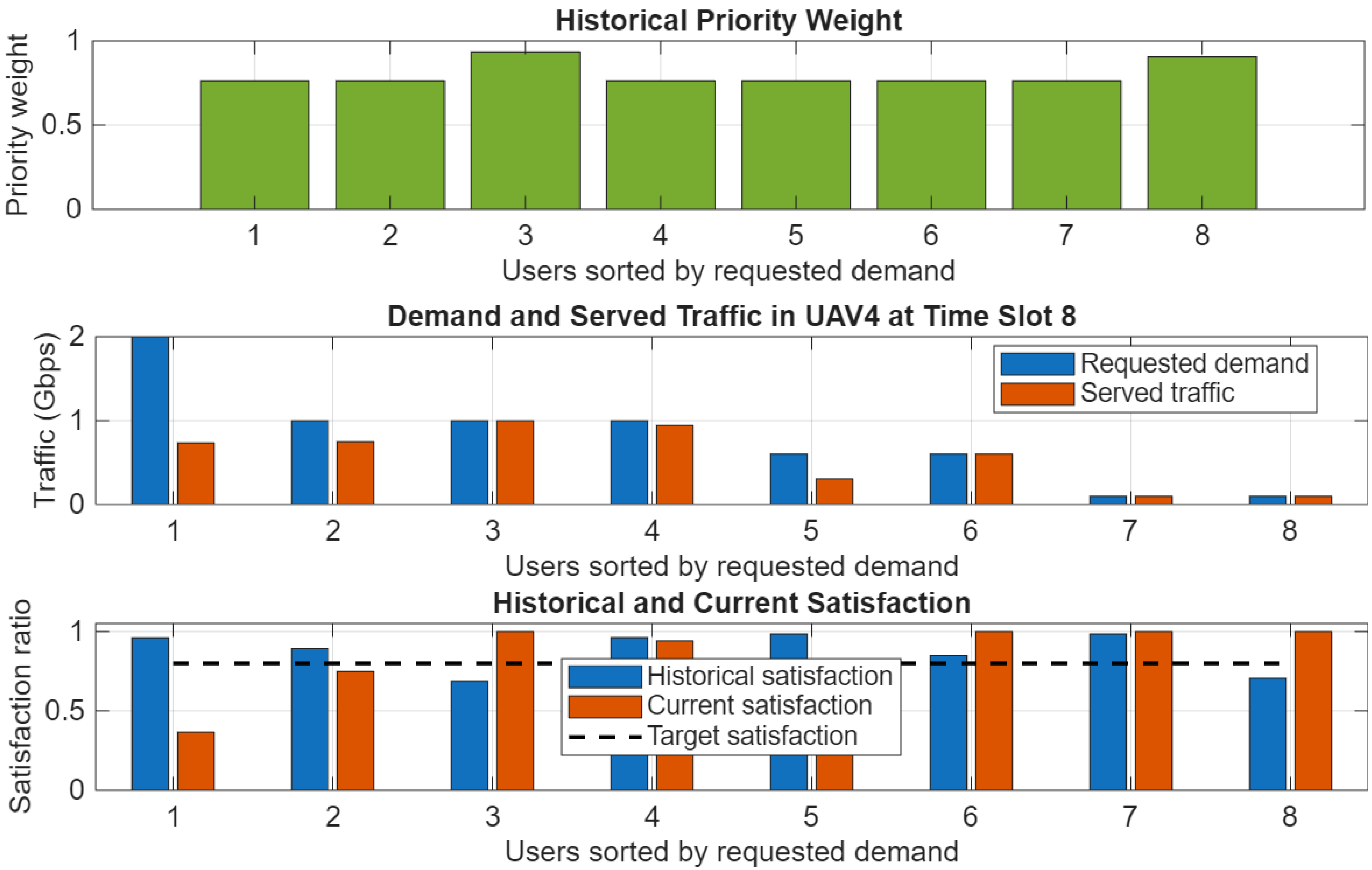
- OIRS blocks are allocated according to the expected improvement in user-level satisfaction, considering backhaul capacity, access capability, and user demand.

5. The Behavior of User 5 in UAV4



- *Timeslot 1: the user has* low satisfaction
- *Timeslot 2: its historical* priority weight increases
- *Timeslot 3: user* reaches the target satisfaction
- *Timeslots 4-5-6:* give priority to other users -> the priority weight gradually decreases
- *Timeslot 7:* satisfaction is decreased again
- *Timeslot 8:* being served

5. The Behavior of Each User in UAV4



- Users with higher priority weights tend to receive more service.

6. Conclusion and Future Improvement

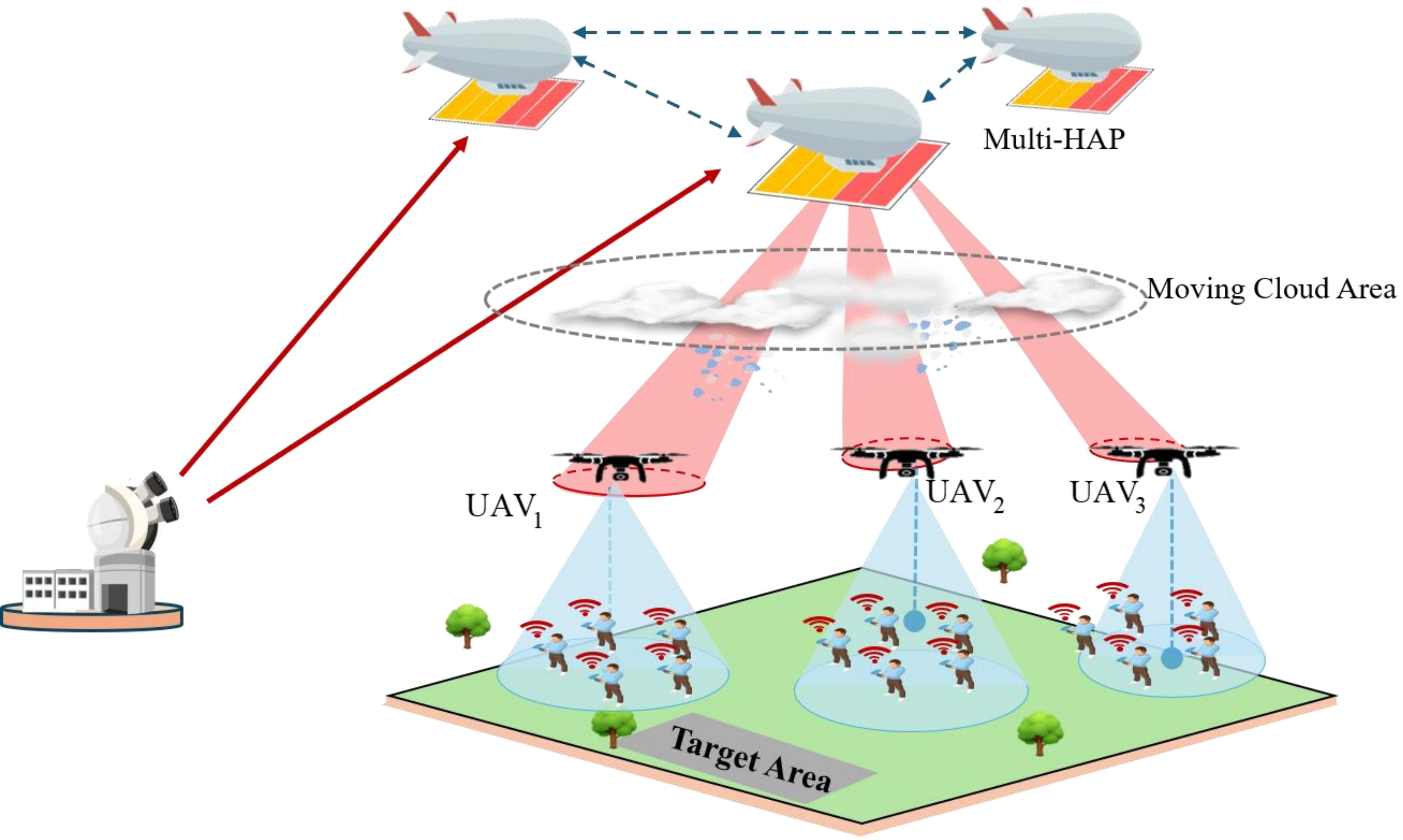
- **In this work, we proposed a *satisfaction-aware resource allocation scheme* for integrated *OIRS-aided HAP and sub-THz access networks*.**

  - The proposed scheme jointly considers OIRS block allocation, sub-THz access scheduling, and historical user satisfaction.

  - The proposed scheme ***improves user-level satisfaction*** and ***satisfied-user coverage***, while showing a throughput-satisfaction tradeoff compared with rate-greedy allocation.

  - Numerical results confirmed the ***effectiveness of proposed scheme*** compared to conventional schemes.

6. Future Improvement: Multi-HAP and Cloud-Aware Availability



- FSO links are sensitive to clouds -> system's outage -> link diversity
- Multi-HAP extension: recent works focus on deploying multi-HAP to avoid clouds, improving outage probability

- Formulate another problem: select the best HAP to guarantee system's availability under moving cloud

$$\mathbf{P1}: \max_{n,p,b,\rho} \sum_{i\in\mathcal{I}} \sum_{k\in\mathcal{K}_i(t)} w_{i,k}(t)\ln(1+\rho_{i,k}(t)) \quad (61a)$$

$$\text{s.t.} \quad \rho_{i,k}(t) = \frac{s_{i,k}(t)}{D_{i,k}^{\text{req}}(t)+\delta}, \quad \forall i\in\mathcal{I},\ k\in\mathcal{K}_i(t), \quad (61b)$$

$$0 \le s_{i,k}(t) \le D_{i,k}^{\text{req}}(t), \quad \forall i\in\mathcal{I},\ k\in\mathcal{K}_i(t), \quad (61c)$$

$$s_{i,k}(t) \le R_{i,k}^{\text{AC}}(\mathbf{p}_i,\mathbf{b}_i;t), \quad \forall i\in\mathcal{I},\ k\in\mathcal{K}_i(t), \quad (61d)$$

$$\sum_{k\in\mathcal{K}_i(t)} s_{i,k}(t) \le R_i^{\text{BH}}(n_i;t), \quad \forall i\in\mathcal{I}, \quad (61e)$$

$$\sum_{k\in\mathcal{K}_i(t)} \sum_{m\in\mathcal{M}} p_{i,k,m}(t) \le P_i^{\text{U}}, \quad \forall i\in\mathcal{I}, \quad (61f)$$

$$\sum_{k\in\mathcal{K}_i(t)} b_{i,k,m}(t) \le B_m, \quad \forall i\in\mathcal{I},\ m\in\mathcal{M}, \quad (61g)$$

$$\sum_{i\in\mathcal{I}} n_i(t) \le N_B, \quad (61h)$$

$$p_{i,k,m}(t)\ge 0, \quad b_{i,k,m}(t)\ge 0, \quad s_{i,k}(t)\ge 0, \quad (61i)$$

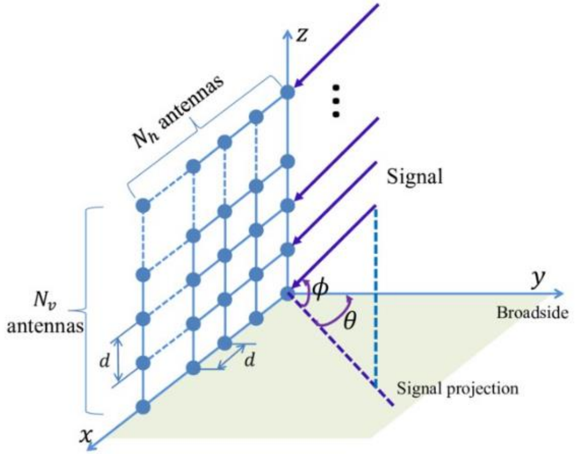
$$n_i(t)\in\mathbb{Z}_+, \quad \forall i\in\mathcal{I}. \quad (61j)$$

- Current optimization problem suffers from the running time due to many variables -> make it difficult for real-time allocation

1. Algorithm: learning-based prediction or use alternative approach

2. System design: uniform planar antenna arrays on UAVs may be consider to support directional sub-THz beams.

   Uniform planar antenna array comprises multiple antennas. Arranging and grouping antenna can create multiple beam forming towards end users.



THANK YOU FOR LISTENING!

References

[1] R. Chen, Y. Sun, L. Liang and W. Cheng, "Joint Power Allocation and Placement Scheme for UAV-Assisted IoT With QoS Guarantee," in IEEE Transactions on Vehicular Technology, vol. 71, no. 1, pp. 1066-1071, Jan. 2022, doi: 10.1109/TVT.2021.3129880.

[2] Z. Phetxomphou, H. D. Le and A. T. Pham, "QoE-Aware UAV Trajectory Planning via Reinforcement Learning for Post-Disaster Network," 2025 IEICE-CS International Conference on Emerging Technologies for Communications

[3] Linh T. Hoang, Chuyen T. Nguyen, Hoang D. Le, and Anh T. Pham, "Adaptive 3D Placement of Multiple UAV-Mounted Base Stations in 6G Airborne Small Cells with Deep Reinforcement Learning," IEEE/ACM Transactions on Networking, vol. 33, no. 4, pp. 1989-2004, Aug. 2025

[4] Thang V. Nguyen, Hoang D. Le, and Anh T. Pham, "On the Design of RIS-UAV Relay-Assisted Hybrid FSO/RF Satellite-Aerial-Ground Integrated Network," IEEE Transactions on Aerospace and Electronic Systems, vol. 59, no. 2, pp. 757-771, Apr. 2023

[5] Khanh D. Dang, Hoang D. Le, Chuyen T. Nguyen, Vuong V. Mai, and Anh T. Pham, "Spatial Resource Allocation for Optical IRS-Aided HAP-Assisted Multi-UAV Networks," IEEE Vehicular Technology Conference, Oslo, Norway, Jun. 2025

[6] S. Zhang and N. Ansari, "3D Drone Base Station Placement and Resource Allocation With FSO-Based Backhaul in Hotspots," in IEEE Transactions on Vehicular Technology, vol. 69, no. 3, pp. 3322-3329, March 2020, doi: 10.1109/TVT.2020.2965920.

[7] D. Wu, X. Sun and N. Ansari, "An FSO-Based Drone Assisted Mobile Access Network for Emergency Communications," in IEEE Transactions on Network Science and Engineering, vol. 7, no. 3, pp. 1597-1606, 1 July-Sept. 2020, doi: 10.1109/TNSE.2019.2942266.

[8] L. Yu, X. Sun, S. Shao, Y. Chen and R. Albelaihi, "Backhaul-Aware Drone Base Station Placement and Resource Management for FSO-Based Drone-Assisted Mobile Networks," in IEEE Transactions on Network Science and Engineering, vol. 10, no. 3, pp. 1659-1668, 1 May-June 2023, doi: 10.1109/TNSE.2022.3233004.

[9] Y. Guan, S. Zou, H. Peng, W. Ni, Y. Sun and H. Gao, "Cooperative UAV Trajectory Design for Disaster Area Emergency Communications: A Multiagent PPO Method," in IEEE Internet of Things Journal, vol. 11, no. 5, pp. 8848-8859, 1 March1, 2024, doi: 10.1109/JIOT.2023.3320796.

[10] S. Shang, E. Zedini, A. Kammoun, and M.-S. Alouini, “A novel hybridoptical and star irs system for ntn communications,” IEEE Trans.Wireless Commun., vol. 25, pp. 3391–3405, 2026

[11] Hung V. Le, Hoang D. Le, Duy-Tuan Dao, and Anh T. Pham, "Multi-Agent Deep Reinforcement Learning for UAV Placement in Optical IRS-Aided Aerial Networks," IEEE Asia-Pacific Conference on Communications, Osaka, Japan, Nov. 2025

[12] Hung V. Le, Hoang D. Le, Duy-Tuan Dao, and Anh T. Pham, "Cloud-Aware Multi-UAV Deployment in FSO-Based Integrated Access and Backhaul Networks via MAPPO," IEEE Transactions on Cognitive Communications and Networking, Apr. 2026

[13] M. Amin Saeidi, H. Tabassum and M. Alizadeh, "Molecular Absorption-Aware User Assignment, Spectrum, and Power Allocation in Dense THz Networks With Multi-Connectivity," in IEEE Transactions on Wireless Communications, vol. 23, no. 11, pp. 16404-16420, Nov. 2024, doi: 10.1109/TWC.2024.3440888.

[14] T. H. do, T. V. Nguyen, H. D. Le, N. T. Dang and A. T. Pham, "DRL-Based Trajectory Design for Laser-Powered UAV-Aided Hybrid FSO/sub-THz Systems," in IEEE Access, vol. 14, pp. 27091-27107, 2026, doi: 10.1109/ACCESS.2026.3665403.